

Chapter 12

DVB Service Information and ATSC Program and System Information Protocol

12.1. INTRODUCTION

The systems part of MPEG-2 provides a range of functions needed to carry MPEG-2 video and audio elementary streams in a single multiplexed bit stream. The *Program-Specific Information (PSI)* in MPEG-2 systems (Chapter 11) allows limited information on the grouping of elementary streams to form programs to be delivered with the elementary streams. Both DVB and ATSC extend this concept, allowing much more side information on the services being delivered to be carried in the stream, each adding a number of its own new tables. In DVB, this additional information is known as *Service Information (SI)*; in ATSC it is known as the *Program and System Information Protocol (PSIP)*.

This chapter begins by examining the reasons why the information carried in the MPEG-2 PSI is not sufficient for a digital television service. This is followed by the operation of the DVB SI and ATSC PSIP. The aim of these sections is to present information on how the SI and the PSIP operate, rather than a complete examination of all the syntax involved. Readers interested in further information will find this in the relevant standards documents. It finishes by analyzing the constraints on the structure of transport streams that can enable embedding both DVB SI and ATSC PSIP in a single stream, allowing decoding by either ATSC or DVB decoders.

12.2. WHY SI AND PSIP?

The MPEG-2 PSI provides sufficient information on the structure of a transport stream for a decoder to identify packets carrying data for each elementary stream, to extract the elementary stream data, and to build complete programs from elementary streams. What is not provided, however, is any information that might be required to extract the transport stream from a distribution network. There is also no information

that might be used to present an electronic program guide or other information on the available programs to the viewer.

The purpose of both the DVB SI and the ATSC PSIP is to extend the functionality of the MPEG-2 PSI in two areas: viewer information and system information. The viewer information may be used by a decoder to provide an electronic program guide. Examples include program titles; cast lists; and information on starting times and program classification, meeting regulatory requirements to indicate the presence of certain types of program content (such as violence).

The system information provides information used internally by a decoder, including the following:

- network information, describing one or more distribution networks;
- conditional access information, providing information to the decoder on any conditional access system in use;
- timing information, passing to the decoder the current time and date; and
- information on services not provided by MPEG-2, such as closed captioning/ subtitling, teletext, and data services.

It is worth noting that, while both DVB and ATSC have been fit to define new tables and descriptors, neither has provided any performance requirements for decoders. Information provided in the profiles and levels for MPEG-2 video makes possible the guarantee that a compliant decoder can successfully decode a compliant bit stream. No such requirements exist for either SI or PSIP.

12.3. DVB-SI

The structure of DVB services is illustrated in Figure 12.1 [1]. A receiver or set-top box receives signals from one or more distribution systems, which are

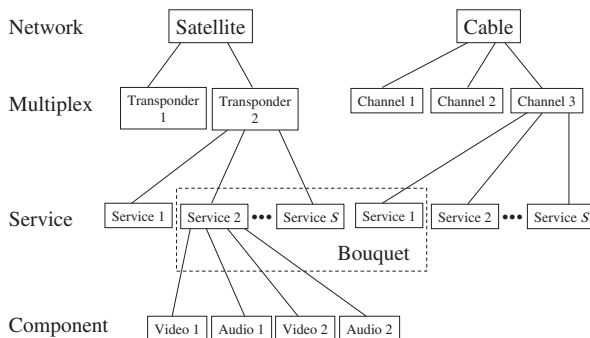


Figure 12.1 DVB service structure. © European Telecommunications Standards Institute 1997. © European Broadcasting Union 2006. Further use, modification, redistribution is strictly prohibited. ETSI standards are available from <http://pda.etsi.org/pda/> and http://www.etsi.org/services_products/freestandard/home.htm.

known as networks in DVB. Possible networks are terrestrial broadcast, cable, satellite, and multipoint microwave distribution systems. Each of these networks delivers one or more multiplexes, each of which is an MPEG-2 system bit stream. Within each multiplexed stream, there are one or more services (corresponding to channels in analog television), each made up of a sequence of one or more programs. Each program consists of a sequence of one or more events. One or more components (corresponding to the MPEG-2 elementary stream) are combined to form each event.

Each network may contain the offerings of more than one service provider. The offerings of a service provider may also be spread across a number of networks. A *bouquet* is a group of services, usually corresponding to the services offered by a single provider. A bouquet may therefore be spread across a number of networks.

DVB SI consists of the four tables defined in the MPEG-2 PSI (program association table (PAT), program map table (PMT), network information table (NIT), conditional access table (CAT)) as well as a number of new tables, known as the *bouquet association table (BAT)*, *service description table (SDT)*, *event information table (EIT)*, *running status table (RST)*, *time and date table (TDT)*, *time offset table (TOT)*, and *stuffing table (ST)*.

The **transport_stream_id** field in the program association table was intended by the designers of MPEG-2 to provide a unique identifier to each transport stream, potentially allowing one transport stream to carry data that refers to another identified transport stream. For ensuring that two network operators do not use the same value of **transport_stream_id**, fixed values can be assigned to each operator. The 16 bits allocated to this field allow 65,536 different streams to be distinguished, which is sufficient to distinguish one network's transport streams but insufficient to identify all transport streams on all networks or to allow permanent allocation of values to network operators. In order to solve this problem, DVB has introduced a new field, the **original_network_id**, carried in the network information table. A fixed assignment of values of the **original_network_id** to network operators is made. Each operator is at liberty to assign values of the **transport_stream_id**. The combination of **original_network_id** and **transport_stream_id** uniquely identifies a transport stream. When a transport stream crosses from one network to another, the value of the **original_network_id** is preserved. The relationship between **network_id**, **original_network_id**, and **transport_stream_id** is illustrated in Figure 12.2.

In addition to carrying SI tables describing itself, a DVB transport stream may also carry tables describing other transport streams. DVB uses the **transport_stream_id** field of the PAT to identify a transport stream described by a table. Tables describing data in the transport stream in which they are carried are said to refer to the *actual transport stream*, whereas the term *other transport stream* is used for tables describing data carried in a different transport stream. Tables referring to another transport stream identify the stream by the combination of its **original_network_id** and **transport_stream_id**. In the case of the network information table, information about the *actual* or *other* networks may also be carried.

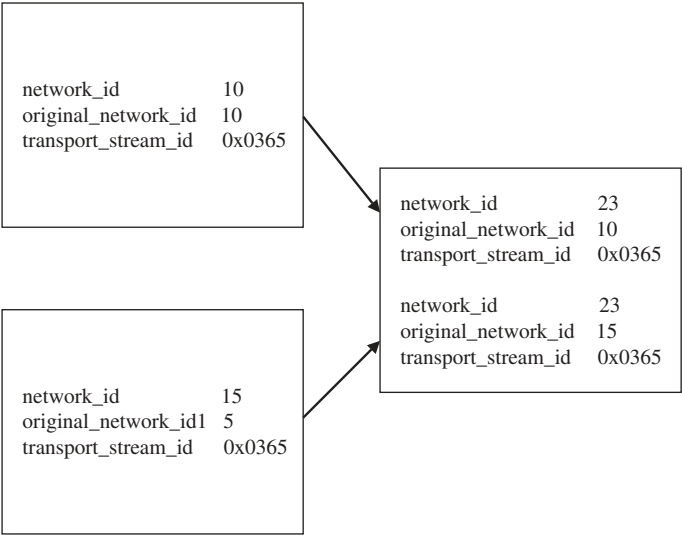


Figure 12.2 Use of network_id, original_network_id, and transport_stream_id to uniquely identify transport streams.

DVB tables are identified in the transport stream using the **table_id** field. DVB also assigns a fixed PID to each table (except for the PMT), which is not allowed to be used for other purposes. Some tables (such as the service description table and the bouquet association table) share PIDs, but have unique values of **table_id**. These PID and **table_id** values are shown in Table 12.1.

Where conditional access is used, tables are not scrambled except for the event information table carrying schedule information (**table_id** = 0x50–0x6F). The event information table may be scrambled because it is often considered that it carries proprietary information. Other tables do carry information that is effective in the public domain, so there is no protection for a network operator in scrambling them.

In DVB a new version of a section of a table becomes effective immediately after the last byte of the 32-bit CRC at the end of the section. Although the syntax supports sections with **current_next_indicator** equal to 0, these are never transmitted.

The decoders in consumer set-top boxes always use the SI data. Professional decoders (used by network operators) may use neither the SI nor the PSI, allowing streams with no SI or unreliable SI to be viewed and possibly debugged.

12.3.1. DVB Common Data Formats

A number of common elements appear in DVB tables and descriptors:

- The code used for text fields is based on the printable values of the ASCII code [2]. The length of text fields is sometimes specified explicitly

Table 12.1 DVB tables.

Table	Abbreviation	PID	table_id
Program association	PAT	0x0000	0x00
Conditional access	CAT	0x0001	0x01
Program map	PMT	Assigned in PAT	0x02
Network information—actual network	NIT	0x0010	0x40
Network information—other network	NIT	0x0010	0x41
Service description—actual TS	SDT	0x0011	0x42
Service description—other TS	SDT	0x0011	0x46
Bouquet association	BAT	0x0011	0x4A
Event information—actual TS present/following	EIT	0x0012	0x4E
Event information—other TS present/following	EIT	0x0012	0x4F
Event information—actual TS schedule	EIT	0x0012	0x50–0x5F
Event information—other TS schedule	EIT	0x0012	0x60–0x6F
Time and date	TDT	0x0014	0x70
Time offset	TOT	0x0014	0x73
Running status	RST	0x0013	0x71
Stuffing	ST	0x0010–0x0014	0x72

immediately prior to the beginning of the field. Otherwise, this length is derived from the value of **descriptor_length** (in a descriptor) or **section_length** (in a table).

- Countries are specified using the three-character (24-bit) codes specified by ISO 3166 [3]. For example, the code for the United Kingdom is “GBR.”
- Languages are specified by the three-character (24-bit) codes specified by ISO 639 [4]. For example, the code for French is “FRE” and the code for English is “ENG.”
- Time is coded as a 24-bit BCD number, specifying hours (*hh*), minutes (*mm*), and seconds (*ss*): *hhmmss*.
- Date is coded as a 16-bit modified Julian date (MJD), calculated as [5]

$$\text{MJD} = 14,956 + D + \lfloor (Y - L - 1900) \times 365.25 \rfloor + \lfloor (M + 1 + L \times 12) \times 30.6001 \rfloor,$$
 where *Y* is the year, *M* is the month number (January = 1, February = 2, etc.), *D* is the day of the month (from 1 to 31), and *L* = 0 if *M* = 1 or *M* = 2 and *L* = 1 otherwise. $\lfloor x \rfloor$ means the integer part of *x*.

- Where the time and date are to be specified together, a 40-bit number is used in which the modified Julian date forms the most significant 16 bits and the time forms the remaining 24 bits.

12.3.2. DVB Descriptors

DVB defines a number of descriptors in addition to those defined by MPEG-2 (discussed in Section 11.6). The properties of these new descriptors, whose syntax follows the generic structure shown in Chapter 11, are shown in Table 12.2. The values of **descriptor_tag** used are all in the range assigned for private descriptors by MPEG-2 (0x40–0xFF). DVB claims values in the range 0x40–0x7F for its own use, leaving the remaining values for user definition.

Table 12.2 Descriptors defined by DVB in EN 300 468.

descriptor_tag	Descriptor	Purpose
0x40	Network name descriptor	Network name in text form.
0x41	Service list descriptor	Specifies a service type for a program number.
0x42	Stuffing descriptor	Allows overwriting of invalidated descriptors without changing size or timing of transport stream or inserting of dummy descriptors as place holders. May be carried in NIT, BAT, SDT, and EIT.
0x43	Satellite delivery system descriptor	Specifies the characteristics of a satellite delivery system.
0x44	Cable delivery system descriptor	Specifies the characteristics of a cable delivery system.
0x45–0x46		Reserved for future use.
0x47	Bouquet name descriptor	Text name of a bouquet.
0x48	Service descriptor	Specifies the name of the service provider and service in text form, along with the service type.
0x49	Country availability descriptor	Specifies countries in which the service is intended to be available and those in which the service is not intended to be available.
0x4A	Linkage descriptor	Specifies a service that can be presented at user request (e.g., additional information related to the service or electronic program guide) or when the selected service is not available (e.g., denied by CA).

Table 12.2 (Continued)

descriptor_tag	Descriptor	Purpose
0x4B	NVOD reference descriptor	Provides a list of services that form a near video-on-demand service, allowing a user to select near to the start of an event by choosing the appropriate service.
0x4C	Time-shifted service descriptor	Used in place of the service descriptor to indicate a time-shifted copy of a service.
0x4D	Short event descriptor	Provides the name of an event and a short text description in a specified language.
0x4E	Extended event descriptor	Detailed text description of an event, including a specification of the language used in the description. Up to 15 of these descriptors may be combined where insufficient space is available in one descriptor.
0x4F	Time-shifted event descriptor	Used in place of the short event descriptor to indicate a time-shifted copy of an event.
0x50	Component descriptor	Identifies the content of an elementary stream (video, audio, or DVB-defined data) and the type of the content (e.g., video 4:3 aspect ratio, audio single channel), and assigns a text name to it.
0x51	Mosaic descriptor	Specifies a mosaic of images for display.
0x52	Stream identifier descriptor	Tags a component to allow association with a component_descriptor.
0x53	CA identifier descriptor	Contains a 16-bit field that identifies the CA system in use. DVB document ETR 162 assigns the values of this field to CA vendors.
0x54	Content descriptor	Classifies content. See example below.
0x55	Parental rating descriptor	Specifies for individual countries the minimum suitable viewing age for service.
0x56	Teletext descriptor	Specifies the language used in the teletext and a magazine and page number.
0x57	Telephone descriptor	Specifies a telephone number for narrow-band dial-up interactive channel.
0x58	Local time offset descriptor	Provides a list of countries and regions with the associated time offsets from coordinated universal time (UTC), expressed in hours and minutes. Also allows for the next time offset and the date of the change to be specified (e.g., beginning or end of summer time) for each region.

(continued)

Table 12.2 (Continued)

descriptor_tag	Descriptor	Purpose
0x59	Subtitling descriptor	Identifies the type of subtitling and the language of the subtitles.
0x5A	Terrestrial delivery system descriptor	Specifies the characteristics of a terrestrial delivery system.
0x5B	Multilingual network name descriptor	Text description of network name in one or more languages.
0x5C	Multilingual bouquet name descriptor	Provides bouquet name in one or more languages.
0x5D	Multilingual service name descriptor	Provides text name of service provider and text description of service in one or more languages.
0x5E	Multilingual component descriptor	Provides a text description of a component in one or more languages.
0x5F	Private data specifier descriptor	Specifies the type of private data, using values listed in ETR 162.
0x60	Service move descriptor	Used to allow a service to be tracked by a receiver from one TS to another TS if it is moved.
0x61	Short smoothing buffer descriptor	A compact version of the MPEG-2 smoothing_buffer_descriptor.
0x62	Frequency list descriptor	Contains a list of center frequencies for a multiplex that is transmitted on more than one frequency.
0x63	Partial transport stream descriptor	Used with partial transport streams; usually associated with digital storage media.
0x64	Data broadcast descriptor	For data broadcast components, identifies the specification to which the broadcast adheres and the language of the data broadcast. An optional text description of the component is also available.
0x65	CA system descriptor	Reserved for definition by the Digital Audio Visual Council (DAVIC).
0x66	Data broadcast id descriptor	Short form of data broadcast descriptor containing only the type of the data broadcast (but without text or language).

Many of these descriptors may appear in more than one table. The scope of the descriptor (i.e., the objects to which it refers) is defined by the table in which it appears, as discussed in Section 12.3.3. The following sections provide a more detailed description of a number of these descriptors.

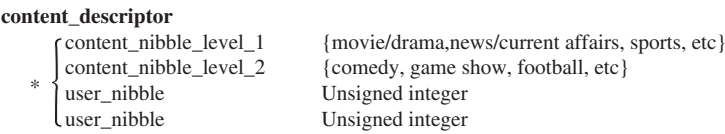


Figure 12.3 Structure of the DVB content descriptor.

12.3.2.1. DVB Content Descriptor

The syntax of the DVB content_descriptor is shown in Figure 12.3, using the simplified syntax defined in Chapter 11. The placement of the bracket and “*” to the left of a field or group of fields indicates that the field or group is repeated zero or more times. This alternative representation is used throughout this chapter.

The body of the DVB content_descriptor consists of four 4-bit fields. The values of the first two of these (**content_nibble_1** and **content_nibble_2**) are standardized by ETSI EN 300 468 [1], allowing a high-level description of the content, such as “drama” or “comedy.” The meanings associated with the remaining two fields (**user_nibble**) is not standardized, but is left for definition by the broadcaster. The mnemonic “uimsbf” means that the field is represented as an unsigned integer and the most significant bit is transmitted first. The purpose of the “for” loop is to allow multiple values to be transmitted for the body of the descriptor. “N” is the number of sets of values contained in the descriptor. This is not transmitted in the bit stream but is inferred from the value of the **descriptor_length** field.

EXAMPLE 12.1—Content Descriptor

The construction of the bit stream for an example content_descriptor is illustrated in Table 12.3. This descriptor specifies a movie/drama (**content_nibble_level_1** = 0x1), of type comedy (**content_nibble_level_2** = 0x4). The values assigned to the two user nibbles are 0x9 and 0xD. The bit stream generated is 0101 0100 0000 0010 0001 0100 1001 1101.

Table 12.3 Sample DVB content descriptor.

Bit stream	Syntax element
0101 0100	descriptor_tag = content_descriptor (0x54)
0000 0010	descriptor_length = 2 bytes
0001	content_nibble_level_1 = movie/drama (0x1)
0100	content_nibble_level_2 = comedy (0x4)
1001	user_nibble = 0x9
1101	user_nibble = 0xD

The alternative representation, avoiding the use of syntax, is shown in Figure 12.4.

content_descriptor	
content_nibble_level_1	movie / drama
content_nibble_level_2	comedy
user_nibble	0x9
user_nibble	0xD

Figure 12.4 Alternative representation of DVB content descriptor example. ■

12.3.2.2. DVB Delivery System Descriptors

DVB provides three descriptors that convey the properties of a particular delivery system: the cable deliver system descriptor, the satellite delivery system descriptor, and the terrestrial delivery system descriptor. All delivery system descriptors have length 13 bytes in order to facilitate swapping of descriptors when a transport stream moves between networks. Without this fixed length, retiming of the transport stream would often be required at network boundaries.

DVB Satellite Delivery System Descriptor The satellite delivery system descriptor carries information on satellite orbital position, modulation, and channel coding. The contents of this descriptor, shown in Figure 12.5, are the frequency (expressed in megahertz, with a resolution of 100 Hz¹), the orbital position (in degrees east or west, with a resolution of 0.1°), the polarization of the transmitted signal (linear vertical/horizontal or circular left/right), the modulation (always QPSK), the symbol rate (in megasymbols (Msym) per second, with a resolution of 100 sym/s), and the coding rate for the inner convolutional coder (1/2, 2/3, 3/4, 5/6, or 7/8). (The operation of the convolutional coder is discussed in Chapter 13.)

satellite_delivery_system_descriptor	
frequency	frequency (in MHz)
orbital_position	degrees east or west
polarization	{linear (vertical/horizontal),circular (left/right)}
modulation	QPSK
symbol_rate	rate in Msym/s
FEC_inner	{ 1/2,2/3,3/4,5/6,7/8 }

Figure 12.5 Simplified syntax for the DVB satellite delivery system descriptor.

EXAMPLE 12.2—Satellite Delivery System Descriptor

An example of the satellite delivery system descriptor, with a frequency of 500.5 MHz, orbital position 150.7° East, vertical polarization, a symbol rate of 10 Msym/s, and a 1/2 rate inner convolutional coder, is shown in Figure 12.6.

¹This could also be expressed as an integer representing the frequency divided by 100 Hz. The description in the main text follows that used by the DVB standards.

satellite_delivery_system_descriptor	
frequency	500.5 MHz
orbital_position	150.6 degrees east
polarization	linear/vertical
modulation	QPSK
symbol_rate	10 Msym/s
FEC_inner	1/2 rate

Figure 12.6 Example satellite DVB delivery system descriptor. ■

cable_delivery_system_descriptor	
frequency	frequency in MHz
FEC_outer	{ 1/2,2/3,3/4,4/5,5/6,7/8 }
modulation	{ 16, 32, 64, 128, 256 } QAM
symbol_rate	rate in Msym/s
FEC_inner	{ 1/2,2/3,3/4,5/6,7/8, none }

Figure 12.7 Structure of the DVB cable delivery system descriptor.

DVB Cable Delivery System Descriptor The cable delivery system descriptor is similar in structure to the satellite delivery system descriptor, as illustrated in Figure 12.7. It carries the frequency (in megahertz, with a resolution of 100 Hz), the coding rates for the inner and outer convolutional coders (1/2, 2/3, 3/4, 5/6, 7/8, or none), and the modulation type (16, 32, 64, 128, or 256 QAM). Modulation and the operation of convolutional coders are discussed in Chapter 13.

EXAMPLE 12.3—Cable Delivery System Descriptor

An example cable delivery system descriptor is shown in Figure 12.8, with frequency 10.5005 GHz, 7/8 rate outer convolution coder, 1/2 rate inner convolutional coder, 64-QAM modulation, and a symbol rate of 20 Msym/s.

cable_delivery_system_descriptor	
frequency	10,500.5 MHz
FEC_outer	7/8 rate
modulation	64-QAM
symbol_rate	20 Msym/s
FEC_inner	1/2 rate

Figure 12.8 Example DVB cable delivery system descriptor. ■

DVB Terrestrial Delivery System Descriptor The terrestrial delivery system descriptor, whose structure is shown in Figure 12.9, carries the center frequency of the channel (in megahertz, with a resolution of 10 Hz), the channel bandwidth (6, 7, or 8 MHz), the constellation (QPSK, 16 QAM, or 64 QAM), a flag indicating the presence or absence of the hierarchical mode, coding rates for the inner convolutional coder in the high- and low-priority streams, the guard interval (1/32, 1/16, 1/8, or 1/4), the transmission mode (2k or 8k), and a flag indicating whether or not other frequencies are in use. The modulation and channel coding used for DVB terrestrial broadcast delivery systems are described in Chapter 13.

terrestrial_delivery_system_descriptor	
center frequency	frequency (in MHz)
bandwidth	bandwidth in MHz
constellation	{ QPSK, 16-QAM, 64-QAM }
hierarchy information	{ Hierarchical, non-hierarchical }
code_rate-HP_stream	{ 1/2, 2/3, 3/4, 5/6, 7/8 }
code_rate-LP_stream	{ 1/2, 2/3, 3/4, 5/6, 7/8 }
guard_interval	{ 1/32, 1/16, 1/8, 1/4 }
transmission mode	{ 2k, 8k }
other_frequency_flag	flag

Figure 12.9 Structure of the DVB terrestrial delivery system descriptor.

EXAMPLE 12.4—Terrestrial Delivery System Descriptor

Figure 12.10 shows the contents of a terrestrial delivery system descriptor, for a system transmitting at 500.5 MHz, with a bandwidth of 7 MHz using 64 QAM in the nonhierarchical mode with a 1/2 rate inner code, a 1/32 guard interval, operating in the 2k mode, and not using other frequencies.

terrestrial_delivery_system_descriptor	
center frequency	500.5 MHz
bandwidth	7 MHz
constellation	64-QAM
hierarchy information	Non hierarchical
code_rate	1/2
guard_interval	1/32
transmission mode	2k
other_frequency_flag	None

Figure 12.10 Example DVB terrestrial deliver system descriptor. ■

12.3.2.3. DVB Name Descriptors

DVB provides a number of descriptors whose function is to carry the name of an object, such as a network, a bouquet, a service, or a component.

DVB Network Name Descriptor The purpose of the network name descriptor is to carry a text name for a network, as illustrated in Figure 12.11. The network name is expressed as a single text string, whose length is limited by the maximum descriptor size of 256 bytes.

network_name_descriptor	
char	String

Figure 12.11 Structure of the DVB network name descriptor.

EXAMPLE 12.5—Network Name Descriptor

An example network name descriptor, carrying the text “News Network” is shown in Figure 12.12.

```
network_name_descriptor
char      "News Network"
```

Figure 12.12 Example DVB network name descriptor. ■

```
bouquet_name_descriptor
char      String
```

Figure 12.13 Structure of the DVB bouquet name descriptor.

DVB Bouquet Name Descriptor The bouquet name descriptor carries text for the name of a particular bouquet in an unspecified language, as illustrated in Figure 12.13. The only data-carrying field of the descriptor is the character string.

DVB Multilingual Network Name Descriptor The multilingual network name descriptor has the same purpose as the network name descriptor, that is, to carry the text name of a network. The multilingual version, however, is able to carry this name expressed in one or more languages, each of which is explicitly identified within the descriptor. The multilingual network name descriptor carries a language identifier (using the format of ISO 639) and a text string for each language in which the network name is specified, as shown in Figure 12.14. The network to which the descriptor refers is not specified in the descriptor, but is inferred from the table in which the descriptor appears.

```
multilingual_network_name_descriptor
* { ISO_639_language_code   ISO 639 language code
    char                   String
```

Figure 12.14 Structure of the DVB multilingual network name descriptor.

EXAMPLE 12.6—Multilingual Network Name Descriptor

A multilingual network name descriptor for a news service is shown in Figure 12.15. Two languages (English and German) are supported, each identified by an ISO 639 language code.

```
multilingual_network_name_descriptor
ISO_639_language_code   English
char                    "News Network"

ISO_639_language_code   German
char                    "Nachrichten"
```

Figure 12.15 Example DVB multilingual network name descriptor. ■

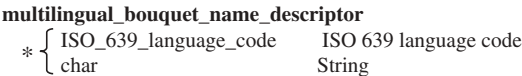


Figure 12.16 Structure of the DVB multilingual bouquet name descriptor.

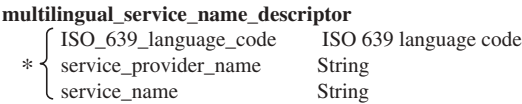


Figure 12.17 Structure of the DVB multilingual service name descriptor.

DVB Multilingual Bouquet Name Descriptor Like the multilingual network name descriptor, the multilingual bouquet name descriptor carries its data in one or more specified languages, as shown in Figure 12.16. The purpose of this descriptor is to carry the name of a bouquet in one or more languages. The bouquet to which the descriptor refers is inferred from the table in which it is carried.

DVB Multilingual Service Name Descriptor The multilingual service name descriptor carries the name of a service provider and service name for one service in one or more specified languages, as shown in Figure 12.17. The service described is not explicitly identified in the descriptor, but is determined from the table in which the descriptor is carried.

DVB Multilingual Component Descriptor The multilingual component descriptor specifies a text name for a component in one or more specified languages, as illustrated in Figure 12.18. The component is identified explicitly in the descriptor by its unique **component_tag**, which appears before the text description. The value of **component_tag** is assigned to a particular elementary stream by placing a stream identifier descriptor in the program map table. The component tag is therefore guaranteed to be unique only within a transport stream.

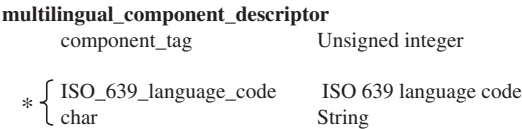


Figure 12.18 Structure of the DVB multilingual component descriptor.

EXAMPLE 12.7—Multilingual Component Descriptor

An example of the multilingual component descriptor is shown in Figure 12.19, for which the component being named has **component_tag** equal to 0x45. The language of the string is English.

```
multilingual_component_descriptor
  component_tag          0x45

  ISO_639_language_code  English
  char                  "English audio"
```

Figure 12.19 Example DVB multilingual component descriptor. ■

```
component_descriptor
  stream_content          {video, audio, teletext, subtitles}
  component_type          {aspect ratio for video, subtitles & teletext,
                           number of channels for audio}
  component_tag           Unsigned integer
  ISO_639_language_code   ISO 639 language code
  text                   String
```

Figure 12.20 Structure of the DVB component descriptor.

12.3.2.4. DVB Component Descriptor

The DVB component descriptor, whose structure is shown in Figure 12.20, describes one component (i.e., elementary stream) present in a transport stream. The **stream_content** tells a decoder that the stream is video, audio, subtitles, or teletext, indicating the appropriate elementary stream decoder to be used. The **component_type** further classifies the stream, specifying for video the aspect ratio and for audio the number of channels. The **component_tag** assigns a unique identifier to the component.

EXAMPLE 12.8—DVB Component Descriptor

An example DVB component descriptor is shown in Figure 12.21. The component described is a video elementary stream with 4:3 aspect ratio, component_tag equal to 0x23, English language, and name “Football-video.”

```
component_descriptor
  stream_content          video
  component_type          4:3 aspect ratio
  component_tag           0x23
  ISO_639_language_code   ENG
  text                   "Football-video"
```

Figure 12.21 Example DVB component descriptor. ■

12.3.2.5. DVB Event Descriptors

DVB uses the term *event* to describe a portion of one program. Two descriptors are available to provide text descriptions of events: the short event descriptor and the extended event descriptor.

DVB Short Event Descriptor The short event descriptor (Fig. 12.22) carries the name of an event and a free-form text description of the event in one specified language. The event to which the descriptor refers is not specified, but is determined from the table in which the descriptor is carried.

short_event_descriptor	
ISO_639_language_code	ISO_639_language_code
event_name_char	String
text_char	String

Figure 12.22 Structure of the DVB short event descriptor.

EXAMPLE 12.9—Short Event Descriptor

An example short event descriptor is shown in Figure 12.23. This descriptor names the event as “Premier League Football,” with a description that includes the names of the teams involved.

short_event_descriptor	
ISO_639_language_code	English
event_name_char	Premier League Football
text_char	Manchester United vs Arsenal

Figure 12.23 Example of the DVB short event descriptor. ■

DVB Extended Event Descriptor The extended event descriptor, which is intended for use in addition to the short event descriptor rather than as a replacement for it, is used to specify additional information about an event (Fig. 12.24). The descriptor begins by specifying the language used in its text, which is followed by one or more pairs of strings that are designed to be presented in a two-column format. The **item_description** carries the string for the left column; the **item_char** carries the data for the right column. This style is useful in an electronic program guide for presenting credits for the program. Further text, for presentation in a single column, can be carried in the **text_char**, which might carry an expanded version of the text in the short event descriptor and might be a general description of the event. The extended event descriptor is unique among descriptors, in that its syntax permits the size of the descriptor to be larger than 256 bytes.

Like the short event descriptor, the event to which the extended event descriptor refers is determined from the table in which it is carried.

extended_event_descriptor	
ISO_639_language_code	ISO_639_language_code
* { item_description_char	String
item_char	String
text_char	String

Figure 12.24 Simplified syntax of the DVB extended event descriptor.

EXAMPLE 12.10—Extended Event Descriptor

An example extended event descriptor is shown in Figure 12.25. The intended, two-column presentation of the **item_description_char** and **item_char** fields is shown in Table 12.4.

extended_event_descriptor	
ISO_639_language_code	English
item_description_char	"Producer"
item_char	"John Smith"
item_description_char	"Director"
item_char	"Hugo Jones"
text_char	"Live from Manchester"

Figure 12.25 Example DVB extended event descriptor.

Table 12.4 Presentation of data from example extended event descriptor of Figure 12.25.

Producer	John Smith
Director	Hugo Jones

12.3.2.6. DVB Descriptors for Ancillary Services

DVB supports several ancillary services, including teletext and subtitles. Each of these ancillary services has an associated descriptor.

DVB Teletext Descriptor The teletext descriptor (Fig. 12.26) specifies for one or more teletext pages, the language, the type of data carried, the magazine number, and the page number. (The operation of ITU-T System B Teletext, which is used in DVB streams, is described in Chapter 14.) A teletext page may be specified as carrying subtitles, as an initial page or additional page, or a program schedule page. Without specification of the page type, the decoder treats a teletext stream in the same way as an analog receiver; use of these fields may support more sophisticated page navigation features. The teletext stream to which the descriptor refers is determined from the placement of the descriptor in the program map table.

teletext_descriptor	
*	ISO_639_language_code ISO_639_language_code
	teletext_type { Subtitle, initial page, additional page, program schedule page }
	teletext_magazine_number Unsigned number
	teletext_page_number Unsigned number

Figure 12.26 Structure of the DVB teletext descriptor.

EXAMPLE 12.11—Teletext Descriptor

An example teletext descriptor is shown in Figure 12.27. Identification of this page as a program schedule facilitates quick presentation, without a requirement for the viewer to explicitly enter the magazine and page number.

teletext_descriptor	
ISO_639_language_code	English
subtitling_type	Program schedule
teletext_magazine_number	5
teletext_page_number	12

Figure 12.27 Example DVB teletext descriptor.

subtitling_descriptor		
*	ISO_639_language_code	ISO_639_language_code
	subtitling_type	{Normal/hearing impaired}
	composition_page_id	Unsigned number
	ancillary_page_id	Unsigned number

Figure 12.28 Structure of the DVB subtitling descriptor.

DVB Subtitling Descriptor The subtitling descriptor is used to identify the type of subtitles (Fig. 12.28). More than one subtitling service may be provided for one program, some supporting services for the hearing impaired while other providing text translation of dialog into other languages. This descriptor is usually carried in the program map table, from which the subtitling stream to which it refers is identified. The values of **composition_page_id** and **ancillary_page_id** refer to the location in the subtitling service where the subtitling data can be found.

DVB subtitling is described in Chapter 14.

EXAMPLE 12.12—Subtitling Descriptor

An example subtitling descriptor is shown in Figure 12.29, in which subtitling in two languages is specified. For English, hearing-impaired subtitles are provided (with a composition page id of 34 and ancillary page id 653), whereas for French, normal subtitles are provided (with a composition page id of 36 and ancillary page id 659). For each type of subtitle, the subtitle page is identified.

subtitling_descriptor	
ISO_639_language_code	English
subtitling_type	Hearing impaired
composition_page_id	34
ancillary_page_id	653
ISO_639_language_code	French
subtitling_type	Normal
composition_page_id	36
ancillary_page_id	659

Figure 12.29 Example of the DVB subtitling descriptor.

12.3.2.7. DVB Service Descriptor

The service descriptor (Fig. 12.30) specifies the type of a service, using one of the values shown in Table 12.5, the name of the service provider, and the name

service_descriptor	
service_type	{ digital television,digital radio,teletext, data broadcast, etc }
service_provider_name	String
service_name	String

Figure 12.30 Structure of the DVB service descriptor.

Table 12.5 Service types supported by the DVB service descriptor.

Service type
Digital television service
Digital radio sound service
Teletext service
NVOD reference service
NVOD time-shifted service
Mosaic service
PAL coded signal
SECAM coded signal
D/D2-MAC
FM radio
NTSC coded signal
Data broadcast service

of the service. The service to which these values refer is inferred from the service description table in which the descriptor is carried. “NVOD” is an abbreviation for near-video-on-demand, which refers to a group of services that are time-shifted from one another to provide a similar service to video-on-demand.

EXAMPLE 12.13—Service Descriptor

An example service descriptor for a news service provided by CNN is shown in Figure 12.31.

service_descriptor	
service_type	digital television
service_provider_name	“CNN”
service_name	“News channel”

Figure 12.31 Example DVB service descriptor. ■

12.3.2.8. DVB Service List Descriptor

The service list descriptor, illustrated in Figure 12.32, provides a means for specifying the types (e.g., digital television, digital radio) of one or more services. It is typically carried in the network information table or bouquet association table. The

service_list_descriptor
* { service_id Unsigned integer
 service_type { digital television, digital radio, teletext, data broadcast, etc }

Figure 12.32 Structure of the DVB service list descriptor.

transport stream in which the service is carried must be inferred from the table in which the service list descriptor is carried. The format of the **service_type** field is the same as in the service descriptor.

EXAMPLE 12.14—Service List Descriptor

Figure 12.33 shows an example of the service list descriptor. The first service, with **service_id** 0x4535, is a digital television service, whereas the second service is a teletext service.

service_list_descriptor
service_id 0x4535
service_type digital television

service_id 0x543D
service_type teletext

Figure 12.33 Example DVB service list descriptor. ■

12.3.2.9. DVB Linkage Descriptor

The linkage descriptor specifies a service that provides extra information about a bouquet, service, or event and has the structure shown in Figure 12.34. The type of the associated information is specified by the **linkage_type** and may be an information service, an EPG service, a CA replacement service, (for use where the conditional access system does not grant access to the service), a service replacement service, or a data broadcast service. The service carrying the associated information is specified by the **transport_stream_id**, **original_network_id**, and **service_id** contained in the descriptor. The bouquet, service, or event to which the associated information is linked is inferred from the table in which the linkage descriptor is transmitted.

linkage_descriptor
transport_stream_id Unsigned integer
original_network_id Unsigned integer
service_id Unsigned integer
linkage_type { information, EPG, CA replacement,
 service replacement, data broadcast }

Figure 12.34 Structure of the DVB linkage descriptor.

EXAMPLE 12.15—Linkage Descriptor

An example linkage descriptor that specifies a service carrying an EPG associated with a service is shown in Figure 12.35.

linkage_descriptor	
transport_stream_id	0xD234
original_network_id	0x5432
service_id	0x4545
linkage_type	EPG

Figure 12.35 Example DVB linkage descriptor. ■

parental_rating_descriptor	
* {	
country_code	ISO-8859 country code
rating	age

Figure 12.36 Structure of the DVB parental rating descriptor.

12.3.2.10. DVB Parental Rating Descriptor

The parental rating descriptor (Fig. 12.36) specifies the minimum age for which a particular content is suitable. A separate specification may be made for each country.

EXAMPLE 12.16—Parental Rating Descriptor

An example parental rating descriptor, specifying a minimum age of 15 for England and 10 for Sweden, is shown in Figure 12.37.

parental_rating_descriptor	
country_code	England
rating	15
country_code	Sweden
rating	10

Figure 12.37 Example DVB parental rating descriptor. ■

12.3.2.11. DVB Local Time Offset Descriptor

The local time offset descriptor is used to specify one or more time offsets from *coordinated universal time* that are used in particular regions of one or more countries. UTC was previously known as *Greenwich mean time (GMT)*. For each region and country, the current local time offset from UTC is specified in hours and minutes, with positive values indicating east and negative values indicating west. The date and time at which the next change in time offset is to occur is also specified (as a modified Julian date), accompanied by the value of this new offset (in hours and minutes). The most likely reason for this offset is a switch to or from summer time (Fig. 12.38).

local_time_offset_descriptor	
* {	
country_code	ISO-3166 country code
region_id	time zone within country
local_time_offset	signed BCD number
time_of_next_change	same format as UTC_time
next_time_offset	signed BCD number

Figure 12.38 Structure of the DVB local time offset descriptor.

EXAMPLE 12.17—Local Time Offset Descriptor

An example local time offset descriptor is shown in Figure 12.39, specifying the time offset for the United Kingdom as plus 1 h (i.e., summer time). The descriptor also specifies a time and date for the next change in this time offset, at which the offset changes to zero.

```
local_time_offset_descriptor
country_code      United Kingdom
region_id        Only one time zone
local_time_offset +0100
time_of_next_change 0xCBD7000000
next_time         +0000
```

Figure 12.39 Example DVB local time offset descriptor. ■

12.3.3. DVB Tables

MPEG-2 defines four tables: the program association table, the program map table, the conditional access table, and the network information table. These tables form a part of the DVB SI. In the case of the program association table, the program map table, and the conditional access table, the full syntax and semantics are defined by MPEG-2. For the network information table, only its existence is defined by MPEG-2; its syntax is defined by DVB. This section discusses special use made by DVB of the MPEG-2 PSI and the new tables introduced by DVB.

12.3.3.1. DVB Use of the Network Information Table

The syntax of the network information table is not defined by MPEG-2 but is defined by DVB, where it is always carried with a PID of 0x0010. Its purpose is to carry information on the physical organization of transport streams on a network and on the network itself. A network information table may describe the network on which it is transmitted (the *actual network*) or another network (*other network*). Transmission of the network information table for the actual network is mandatory.

One section of the network information table (Fig. 12.40) is required for each network to be described. The network information section consists of a unique network identifier (assigned in ETR 162 [6]), a group of descriptors describing this network, and a list of transport streams carried by the network, each of which may be accompanied by one or more descriptors describing that transport stream. Each

```
network_information_section-(actual/other)_network
network_id      unsigned integer
* << descriptors >>

* {
  transport_stream_id      unsigned integer
  original_network_id      unsigned integer
  << descriptors >>
}
```

Figure 12.40 Simplified syntax of the DVB network information table.

of these transport streams is identified by its combination of **transport_stream_id** and **original_network_id**.

The network information table is not scrambled.

A number of descriptors have meaning when carried in the network information table, some describing the network as a whole and others describing one multiplex carried by the network. Descriptors that can be used to describe the network to which the table refers are the linkage descriptor, the network name descriptor, and the multilingual network name descriptor. The linkage descriptor is used to point to services associated with the network as a whole, such as electronic program guides or network information. The network name descriptor and multilingual network name descriptor carry a text form of the network’s name, such as “BSkyB,” or “ASTRA.”

Descriptors carried by the network information table that describe individual multiplexes are the delivery system (cable, satellite, and terrestrial) descriptors, the service list descriptor, and the frequency list descriptor. The carriage of one delivery system descriptor describing each multiplex is required. The service list descriptor may contain a list of services carried by the multiplex, identified by their **service_id**, which is the same as the MPEG-2 **program_number**. The frequency list descriptor may carry a list of other frequencies used in transmission of the multiplex.

EXAMPLE 12.18—DVB Network Information Table

An example network information table is shown in Figure 12.41. This network has a network identifier of 0x0069 and a network name “Satellite Network.” This network carries two

network_information_section-actual_network	
network_id	0x0069
network_name_descriptor	
char	“Satellite Network”
transport_stream_id	0x0234
original_network_id	0x0001
satellite_delivery_system_descriptor	
frequency	12.750 MHz
orbital_position	156.0° E
bandwidth	7 MHz
polarization	linear/vertical
modulation	QPSK
symbol_rate	14 Msym/s
FEC_inner	1/2
transport_stream_id	0x0239
original_network_id	0x0001
satellite_delivery_system_descriptor	
frequency	12.720 MHz
orbital_position	156.0° E
bandwidth	10 MHz
polarization	linear/vertical
modulation	QPSK
symbol_rate	20 Msym/s
FEC_inner	3/4

Figure 12.41 Example DVB network information table.

multiplexes, each of which is identified by its unique combination of **transport_stream_id** and **original_network_id**. Each of these multiplexes is described by a satellite delivery system descriptor. ■

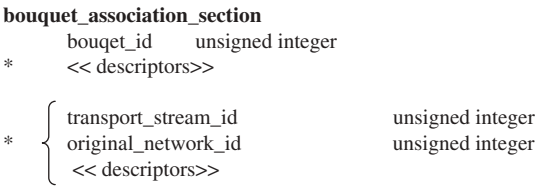


Figure 12.42 Simplified syntax for the DVB bouquet association table.

12.3.3.2. DVB Bouquet Association Table

The purpose of the bouquet association table is to provide information on bouquets, which are collections of services, possibly crossing network boundaries. The bouquet association table has the same structure as the network information table. Whereas the network information table provides information on the physical arrangement of multiplexes and services, the bouquet association table describes the logical grouping of services into bouquets.

Each section of the bouquet association table, whose syntax is illustrated in Figure 12.42, contains a **bouquet_id** that uniquely identifies the bouquet to which the table refers. These values are assigned in ETR 162. BSKyB, for example, is assigned values in the range 0x1000–0x101F. The **bouquet_id** is followed by a list of descriptors describing the bouquet as a whole. These descriptors may include the bouquet name descriptor and multilingual bouquet name descriptor, CA identifier descriptor, country availability descriptor, and linkage descriptor.

The bouquet name descriptor and the multilingual bouquet name descriptor may be used in the bouquet association table to provide a text name for the bouquet. The CA identifier descriptor identifies conditional access used by services within the bouquet. The country availability descriptor specifies countries in which the entire bouquet is to be available or not. The linkage descriptor is used to specify a service that carries information on the bouquet or an EPG.

For each transport stream carrying services belonging to a bouquet, the bouquet association table carries the **transport_stream_id**, **original_network_id**, and a service list descriptor that associates one or more services in that transport stream with the bouquet.

EXAMPLE 12.19—Bouquet Association Table

An example bouquet association table is shown in Figure 12.43. This bouquet is spread across two transport streams, encompassing two services on each (giving a total of four services).

bouquet_association_section	
bouquet_id	0x1000
bouquet_name_descriptor	
char	"Satellite Bouquet"
transport_stream_id	0x0234
original_network_id	0x1056
service_list_descriptor	
service_id	0x4520
service_type	digital television
service_id	0x5123
service_type	teletext
transport_stream_id	0x028F
original_network_id	0x1056
service_list_descriptor	
service_id	0x4535
service_type	digital television
service_id	0x543D
service_type	digital radio

Figure 12.43 Example DVB bouquet association table. ■

12.3.3.3. DVB Service Description Table

The service description table defines the services carried by a network. A service is similar in concept to an analog television channel and to the MPEG-2 concept of a program.

Each section of the service description table refers to one transport stream, identified by its **transport_stream_id** and its **original_network_id**. For each service present in the transport stream, the section carries the **service_id**, flags to indicate the presence of event information tables for the schedule and the present and following events, and a flag to indicate the use of conditional access in one or more components of the service. The **service_id** uniquely identifies a service within a transport stream. A service can be uniquely identified among all services on all networks using its **service_id** with the **transport_stream_id** and **original_network_id** of its host transport stream. The **program_number** in the program map table has the same value as the **service_id**. It is recommended by the DVB standards that a fixed **service_id** (and therefore **program_number**) be allocated to each service so that a decoder can store a list of previously used services to allow the viewer to return easily (Fig. 12.44).

Each descriptor carried by the service description table refers to the one service with whose **service_id** it is grouped. Descriptors that are likely to be carried in this table include the service descriptor (specifying the name of the service and the service operator) and multilingual service descriptor, the bouquet name descriptor, the CA identifier descriptor, the country availability descriptor, the data broadcast descriptor, the linkage descriptor (used to specify services that contain additional information about this service, that carry an EPG for this service, or that is a replacement service for this service if it is unavailable), the mosaic descriptor, the

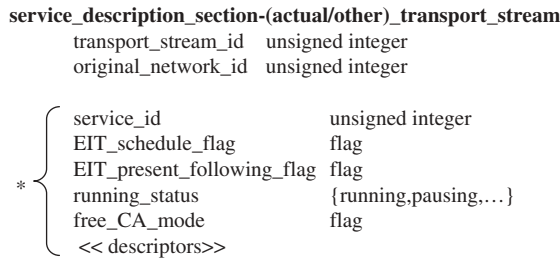


Figure 12.44 Simplified syntax for the DVB service description table.

NVOD reference descriptor, the telephone descriptor, and the time-shifted service descriptor.

The **EIT_schedule_flag** and **EIT_present_following_flag** indicate the presence (or absence) of the event information table for the schedule and the present/following events, respectively. The **running_status** is used to indicate whether a program is currently running, paused, or about to run. The **free_CA_mode** field is set to one for services in which conditional access is not used on any component.

EXAMPLE 12.20—Service Description Table

An example service description table is shown in Figure 12.45, carrying information on two digital television services carried in the same transport stream as the table. For each of these services, event information tables are present for both the schedule and the present/following events, and no conditional access is used. Both services are running.

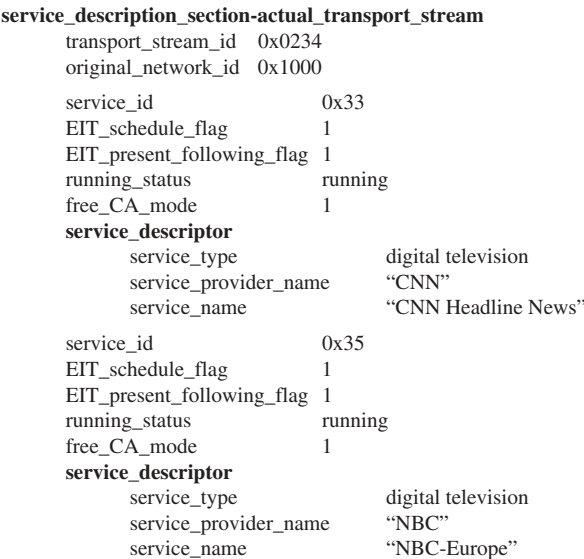


Figure 12.45 Example DVB service description table. ■

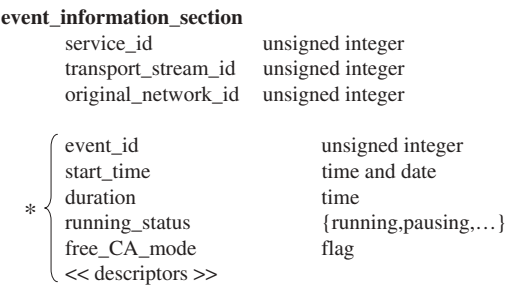


Figure 12.46 Simplified syntax for the DVB event information table.

12.3.3.4. DVB Event Information Table

The purpose of the event information table is to provide information in chronological order on the events within each service. Each section of the event information table contains information on one service, which is identified by its **service_id** and the **transport_stream_id** and **original_network_id** of the transport stream carrying the service (Fig. 12.46).

Unlike other tables, the event information table may be scrambled.

Each descriptor carried by the event information table refers to one event. Descriptors that are likely to be present include the component descriptor, the content descriptor, the data broadcast descriptor, the extended event descriptor, the linkage descriptor, the multilingual component descriptor, the parental rating descriptor, the short event descriptor, the telephone descriptor, and the time-shifted event descriptor.

EXAMPLE 12.21—Event Information Table

An example event information table is shown in Figure 12.47, showing information on two events (numbered 0x32 and 0x33) associated with the **service_id** 0x31. ■

12.3.3.5. DVB Running Status Table

The purpose of the running status table is to enable fast updating of the timing status of one or more events, which may be necessary when the starting time of an event is altered. This might occur when the preceding event finishes later or earlier than programmed. The running status of an event is one of the following values: “undefined,” “not running,” “starts in a few seconds,” “pausing,” or “running.” The running status table is not scrambled. The running status table is not able to carry any descriptors.

EXAMPLE 12.22—Running Status Table

An example running status table is shown in Figure 12.48. This table contains entries for three events. The first two share the same transport stream (because they have the same value of **transport_stream_id** and **original_network_id**). The third entry refers to a dif-

event_information_section

service_id 0x31
 transport_stream_id 0x0300
 original_network_id 0x1020

event_id 0x32
 start_time 0xCBD7000000
 duration 3600
 running_status running
 free_CA_mode 1

short_event_descriptor

ISO_639_language_code English
 event_name_char Premier League Football
 text_char Manchester United vs Arsenal

event_id 0x33
 start_time 0xCBD7000100
 duration 1800
 running_status running
 free_CA_mode 1

short_event_descriptor

ISO_639_language_code English
 event_name_char BBC News
 text_char BBC early edition news service

parental_rating_descriptor

country_code England
 rating 5

country_code Sweden
 rating 3

Figure 12.47 Example DVB event information table.

ferent transport stream. The only means to determine if a reference in the running status table refers to a service in the same transport stream is to compare the transport stream's values of **transport_stream_id** and **original_network_id** to those in the running status table.

running_status_section

transport_stream_id 0x0300
 original_network_id 0x1020
 service_id 0x20
 event_id 0x54
 running_status Pausing

transport_stream_id 0x0300
 original_network_id 0x1020
 service_id 0x31
 event_id 0x65
 running_status Running

transport_stream_id 0x1000
 original_network_id 0x17F1
 service_id 0xF4
 event_id 0x45
 running_status Starts in a few seconds

Figure 12.48 Example DVB running status table. ■

program time_and_date_section
UTC_time number

Figure 12.49 Simplified syntax of the DVB time and data table.

12.3.3.6. DVB Time and Date Table

The purpose of this table is to carry time and date information, with the syntax illustrated in Figure 12.49. Time is expressed as a six-digit number (hh:mm:ss), coded as BCD. The date is expressed as a modified Julian date and is coded as a 16-bit binary number. This table is always transmitted as a single section and is not scrambled. It is not able to carry any descriptors. The **UTC_time** field is 40 bits in length, with the most significant 16 bits corresponding to the least significant 16 bits of the MJD, and the remaining 24 bits carrying the six BCD digits for time.

EXAMPLE 12.23—Time and Date Table

The time 08:05:10 (i.e., 5 min and 10 s after 8 a.m.) is represented in BCD as 080510. The date “September 21, 2001” corresponds to $D = 21$, $M = 9$, $Y = 2001$, and $L = 1$, for which MJD = 52,173, or 0xCBCD. The contents of the time and date table for this example are shown in Figure 12.50.

time_and_date_section
UTC_time 0xCBCD080510

Figure 12.50 Example DVB time and date table. ■

12.3.3.7. DVB Time Offset Table

The purpose of the time offset table is to carry the current time and date information. Time and date information is carried in UTC. The table can also carry a list of countries and regions and their associated time offsets, allowing conversion of time and data to local values. This table is not scrambled and is always transmitted as a single section.

The header of the time offset table contains the current time (in UTC, coded in the same manner as that for time and date table) and date (expressed as a modified Julian date, coded as that for the time and date table). The local time offset descriptor is typically the only descriptor carried in this table (Fig. 12.51).

program time_offset_section
UTC_time number
<< descriptors >>

Figure 12.51 Simplified syntax for the DVB time offset table.

EXAMPLE 12.24—Time Offset Table

Figure 12.52 shows an example time offset table showing the same time and date as in the previous example of the time and date table. This time offset table also contains a local time offset descriptor, which specifies information for two countries: the United Kingdom and Australia. For the United Kingdom, the **region_id** shows that only one time zone is used. The current

local_time_offset is plus 1 h, which will change to zero at midnight on October 1, 2001. For Australia, the **region_id** specifies the eastern-most time zone, where the current time offset is plus 10 h, which will change to plus 11 h at 2 a.m. on October 7, 2001.

time_offset_section	
UTC_time	0xCBCD080510
local_time_offset_descriptor	
country_code	United Kingdom
region_id	Only one time zone
local_time_offset	+0100
time_of_next_change	0xCBD7000000
next_time	+0000
country_code	Australia
region_id	Eastern time zone
local_time_offset	+1000
time_of_next_change	0xCBDE020000
next_time	+1100

Figure 12.52 Example DVB time offset table. ■

12.3.3.8. DVB Stuffing Table

The purpose of the stuffing table is to allow a section becoming invalid as a transport stream crosses a network boundary to be overwritten with null data. This means that the data rate of the transport stream remains constant. A stuffing table may also be inserted in anticipation of a section being added when the transport stream crosses a network boundary. This table is not scrambled and contains no descriptors. All bytes after the header have the value 0xFF.

12.3.4. DVB Delivery Issues

12.3.4.1. Size of Tables

Although tables larger than 1024 bytes are supported by the DVB standard, the maximum size of each table section is limited to 1024 bytes. A table may be sufficiently long so that it requires two or more sections for transmission simply because an encoder places a large number of descriptors in the table. This may also occur because of the use of a number of free-text descriptors whose length is not constrained.

When a TS is remultiplexed, new descriptors may be added to one or more tables. Some descriptors may also be deleted. When descriptors are deleted, the system encoder may shorten the table. Alternatively, it may simply overwrite the deleted descriptor with a stuffing descriptor. This use of stuffing descriptors can lead to a growth in the length of tables, especially when a number of stages of remultiplexing occur. This growth in the length of tables is supported by the standard through the use of multisection tables. Many encoders, however, do not support multisection tables. Use of these encoders in the remultiplexing of streams that contain multisection tables can lead to corruption of SI data.

Table 12.6 Maximum intervals in seconds between retransmission of tables.

Table	Satellite/Cable	Terrestrial
NIT	10	10
BAT	10	10
SDT	Actual TS: 2 Other TS: 10	Actual TS: 2 Other TS: 10
TDT	30	30*
TOT	30	30*
EIT (Present/following)	Actual TS: 2 Other TS: 10	Actual TS: 2 Other TS: 10
EIT (Schedule Table)	First 8 days: 10 Later: 30	Actual TS First 8 days: 10* Later: 30* Other TS First 8 days: 60* Later: 300*

Note: entries marked “*” are recommended. All other entries are mandatory.

12.3.4.2. Table Entries

To be decoded, a service must contain one or more elementary streams (containing video, audio, or private data), an entry in the PAT for a PMT, and the PMT specified in the PAT. A service missing any one of these cannot be decoded. A DVB bit stream may, however, contain elementary streams that do not have corresponding PAT and PMT entries. There may also exist services that are defined by the SI but with no PSI or elementary streams, which are sometimes referred to as *blank programs*. Blank programs occur when SI for a service is carried in a transport stream separate from the one carrying the service itself.

12.3.4.3. Repetition Rates

The DVB standard defines the maximum intervals between transmission for each type of table, shown in Table 12.6 [7]. Regular retransmission of tables minimizes the impact of propagation errors on the decoded service and the time taken to begin decoding a transport stream. Different repetition rates are specified for satellite, cable, and terrestrial broadcast systems.

In a DVB system, approximately 5% of the network capacity is allocated to the transmission of SI tables and CA information. In a 40 Mbit/s satellite distribution system, the total SI and CA data is likely to be 2 Mbit/s.

12.4. ATSC PROGRAM AND SYSTEM INFORMATION PROTOCOL

ATSC’s program and the PSIP provides a small collection of tables and associated descriptors that can be used to describe the contents of one or more transport streams. These

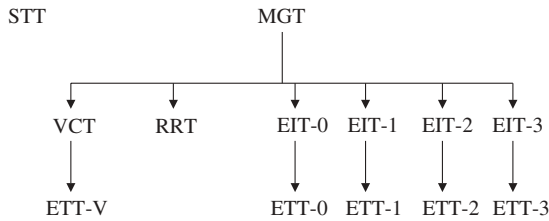


Figure 12.53 Logical structure of ATSC PSIP tables.

tables can also be used to describe the contents of analog transmissions. The purpose of ATSC PSIP [8] is similar to that of DVB SI, although it is much simpler in structure.

ATSC continues to use the program association table, program map table, and conditional access table defined by MPEG-2. In addition, ATSC defines

- a *system time table (STT)*, used to specify the current UTC time;
- a *master guide table (MGT)*, which provides an index to the other tables;
- a *virtual channel table (VCT)*, which describes the virtual channels that are carried by one or more transport streams and also provides the facilities of the network information table;
- a *rating region (RRT)* table that conveys information on the program classification system in use in one or more regions;
- *event information tables*, which describe the upcoming programs on all virtual channel defined in the virtual channel table; and
- the *extended text table*, which provides text descriptions that can be used to augment the virtual channel table or event information table.

The logical structure of these tables is shown in Figure 12.53, where the master guide table sits at the top of the hierarchy, with the virtual channel table, rating region table, and event information tables below it. The virtual channel table and event information tables may each have a child extended text table. The system time table sits outside this hierarchy.

In this section, we begin with an overview of the common data formats used by the ATSC's PSIP. This is followed by a description of each of the descriptors and tables.

12.4.1. Common Data Formats

ATSC uses common data formats for the representation of strings and time and date.

12.4.1.1. String Representation

ATSC uses the multiple string structure, illustrated in Figure 12.54, as a common means for specifying strings in one or more identified languages. (The format for the structure is set out in Section 12.3.2.1.) The language is specified by an ISO 639 three-character code. Strings may be uncompressed, or compressed using one of the

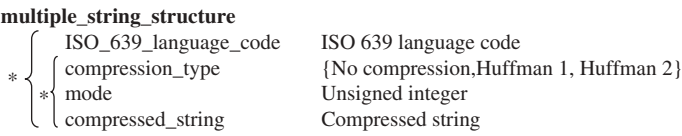


Figure 12.54 Structure of the ATSC multiple string structure.

two Huffman codes. The character set is based on an 8- or 16-bit code defined by ISO 10646, with the page number controlled by the **mode** field.

EXAMPLE 12.25—Multiple String Structure

Figure 12.55 shows an example multiple string structure carrying the string “Hello” in English and in French. In both cases the mode has value 0x00, the ISO 10646 value indicating the Latin-1 alphabet.

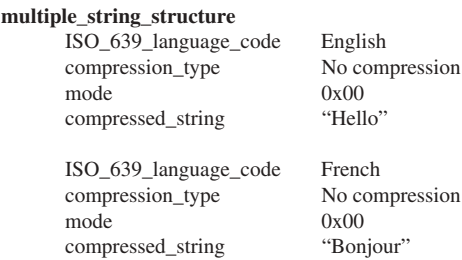


Figure 12.55 Example ATSC multiple string structure. ■

12.4.1.2. ATSC Time and Date Formats

Time and date in ATSC tables is specified as a 32-bit unsigned integer representing the number of seconds since midnight, January 6, 1980, based on the GPS time reference. This GPS time is translated to UTC time by adding an offset carried by the system time table.

EXAMPLE 12.26—ATSC Time and Date

The GPS time at 12:00, August 7, 2001, is 681220800. ■

12.4.1.3. ATSC Virtual Channels

Each ATSC transport stream may carry several programs simultaneously. A means is required to identify uniquely each of these programs and possibly to identify programs in other transport streams. ATSC uses virtual channel numbers for this purpose, with each virtual channel number identifying a particular program that the viewer can select. Each virtual channel is uniquely identified by a major channel

Table 12.7 ATSC Descriptors.

descriptor_tag	Descriptor	Purpose
0x80	Stuffing descriptor	Acts as a place holder in a table.
0x81	AC-3 audio descriptor	Describes an AC-3 audio elementary stream.
0x86	Caption service descriptor	Carries characteristics of closed caption services associated with an event.
0x87	Content advisory descriptor	Conveys advisory indications of program classification for one or more regions.
0xA0	Extended channel name descriptor	Carries the long form of the name of a virtual channel.
0xA1	Service location descriptor	Specifies the stream types, PID, and language for each elementary stream.
0xA2	Time-shifted service descriptor	For one virtual channel, identifies other virtual channels that are time-shifted copies.
0xA3	Component name descriptor	Provides a text name for one component.
0xC0–0xFE	User private	Available for definition by users without interference with the operation of ATSC

number and a minor channel number. The major channel number is assigned based on the RF channel number on which the operator broadcasts. For broadcasters who already hold an NTSC license, the major channel number is the number of this channel. For new, digital-only broadcasters, the major channel number is the number of their assigned digital channel. This method of assignment means that, in a given geographic area, only one major channel number is used by each broadcaster, even if that broadcaster controls more than one RF channel. Minor channel numbers are used to identify individual virtual channels.

12.4.2. ATSC Descriptors

ATSC defines a number of descriptors in addition to those defined by MPEG-2 (Chapter 11), whose characteristics are summarized in Table 12.7, to complement those defined in MPEG-2 systems. The values of **descriptor_tag** assigned by ATSC are in the range 0x80–0xAF, which are values left unassigned by MPEG-2. ATSC restricts user-defined descriptor tags to the range 0xC0–0xFE.

12.4.2.1. ATSC AC-3 Audio Descriptor

The AC-3 audio descriptor [9] is used to provide information about an AC-3 audio elementary stream. It is usually transmitted in the program map table, with the structure shown in Figure 12.56.

The **bsid** and **bsmod** fields take the same values as the fields of the same name in the AC-3 audio elementary stream. The **bit_rate_code** specifies either the exact bit rate or the upper limit for the bit rate for values between 32 and 640 kbit/s. The **num_channels** field specifies either the number of audio channels encoded in the

AC3_audio_descriptor	
sample_rate_code	{48,44.1,32} kHz
bsid	bit string
bit_rate_code	{32-640} Kbps
surround_mode	{Not indicated, not Dolby surround, Dolby surround}
bsmod	bit string
num_channels	bit string
full_svc	flag
langcod	bit string
text	String

Figure 12.56 Structure of the ATSC AC-3 audio descriptor.

AC-3 elementary stream or an upper bound on the number of channels. The **langcod** field is set to the same value as the langcode field in the AC-3 elementary stream, and represents the language of the audio. A value of zero is used to indicate an unknown language. A value of zero for **full_svc** indicates that this audio elementary stream is not suitable for presentation on its own but should be combined with some other audio service. This might occur where an audio elementary stream contains only special effects and is designed to be combined with another containing dialog. The use of surround sound is indicated by the **surround_mode** field. The **text** field can be used to carry a text description of the audio elementary stream and can be encoded using either the 8-bit ASCII or the 16-bit Unicode formats. Unlike other text strings in ATSC descriptors, this field does not use the multiple string structure.

12.4.2.2. ATSC Caption Service Descriptor

The caption service descriptor (Fig. 12.57) conveys the characteristics of closed captioning that are associated with an event. The captioning systems used in ATSC are described in Chapter 14. Up to 16 different closed captioning services can be associated with one event, for each of which the language, its type (ATVCC or Line 21), the transmission field for Line 21 data, and the caption service number for ATVCC captions, and whether or not the captions are specially formatted for beginner readers or for a wide aspect ratio is specified.

caption_service_descriptor		
*	language	ISO 639 language code
	cc_type	{ ATVCC,Line 21 } Captions
	if cc_type == 0	
	line21_field	{ Field 1, Field 2 }
	else	
	caption_service_number	Unsigned integer
	easy_reader	flag
	wide_aspect_ratio	flag

Figure 12.57 Structure of the ATSC caption service descriptor.

EXAMPLE 12.27—Caption Service Descriptor

An example caption service descriptor is shown in Figure 12.58, specifying two ATVCC closed captioning services. The first is in English, using the caption service number 4 and is aimed at beginner readers. The second service is in French and uses the caption service number 6.

caption_service_descriptor	
language	English
cc_type	ATVCC Captions
caption_service_number	4
easy_reader	1
wide_aspect_ratio	0
language	French
cc_type	ATVCC Captions
caption_service_number	6
easy_reader	0
wide_aspect_ratio	0

Figure 12.58 Example ATSC caption service descriptor. ■

content_advisory_descriptor				
*	{	rating_region	Unsigned integer	
		*	rating_dimension_j	Unsigned integer
			rating_value	Unsigned integer
		rating_description_text	Multiple string structure	

Figure 12.59 Structure of the ATSC content advisory descriptor.

12.4.2.3. ATSC Content Advisory Descriptor

The content advisory descriptor, carried in the rating region table, is used to indicate program classifications for up to eight regions for each event. The region rating table (Section 12.4.3.5) defines one or more dimensions of rating (e.g., language, sexual content) and one or more levels of content for each dimension. The content advisory descriptor presents a list of dimensions and the corresponding rating values for a particular region (Fig. 12.59).

12.4.2.4. ATSC Extended Channel Name Descriptor

The extended channel name descriptor provides the long channel name for a virtual channel. Its payload carries one multiple string structure (Fig. 12.60).

12.4.2.5. ATSC Service Location Descriptor

The service location descriptor specifies the stream types, PID, and language for each elementary stream. The Valid values for the **stream_type** field include MPEG-2 video and AC-3 audio. One service location descriptor is carried in the virtual channel table for each active channel and refers to the elementary streams associated with the current event on that channel (Fig. 12.61).

extended_channel_name_descriptor	
long_channel_name_text	Multiple string structure

Figure 12.60 Structure of the ATSC extended channel name descriptor.

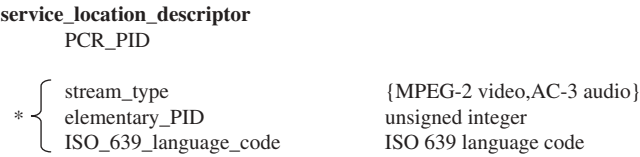


Figure 12.61 Structure of the ATSC service location descriptor.

EXAMPLE 12.28—Service Location Descriptor

An example service location descriptor is shown in Figure 12.62 for an event with two associated elementary streams, one containing MPEG-2 video and the other containing AC-3 audio. For each elementary stream, the language is specified as English. The PCR is carried in the transport stream packets with the same PID as the video elementary stream.

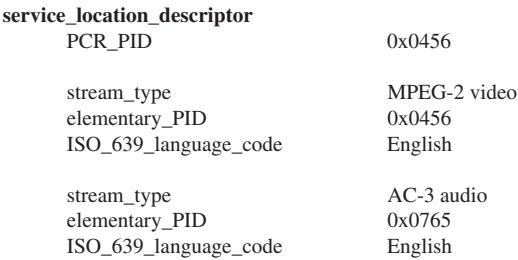


Figure 12.62 Example ATSC service location descriptor. ■

12.4.2.6. ATSC Time-Shifted Service Descriptor

The time-shifted service descriptor, carried in the virtual channel table, links one virtual channel with one or more other virtual channels that carry the same content at a different time. NVD services are the most likely application on this descriptor. As illustrated in Figure 12.63, for each linked virtual channel, this descriptor carries the time shift in minutes, and the major and minor channel numbers of the linked virtual channel. Up to 20 linked virtual channels may be specified.

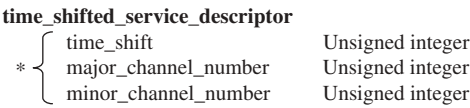


Figure 12.63 Structure of the ATSC time-shifted service descriptor.

EXAMPLE 12.29—Time-Shifted Service Descriptor

An example time-shifted service descriptor is shown in Figure 12.64, showing three linked virtual channels. The major channel number of each of the linked virtual channels is 12. The first linked virtual channel is time shifted by 60 min and has minor channel number 4, the second is time shifted by 30 min with a minor channel number of 5, and the third is time shifted by 90 min and has a minor channel number of 6.

time_shifted_service_descriptor	
time_shift	30
major_channel_number	12
minor_channel_number	4
time_shift	60
major_channel_number	12
minor_channel_number	5
time_shift	90
major_channel_number	12
minor_channel_number	6

Figure 12.64 Example ATSC time-shifted service descriptor. ■

12.4.2.7. ATSC Component Name Descriptor

The component name descriptor carries a single multiple string structure, which can be used as the text name of one component, such as a single video or audio elementary stream (Fig. 12.65).

12.4.2.8. ATSC Stuffing Descriptor

The stuffing descriptor may appear in any ATSC table and may be used as a place holder where it is known that it may be necessary to insert a new descriptor, for example, when the transport stream crosses a network boundary. The stuffing descriptor may also be used to replace a descriptor that is removed from a transport stream, avoiding the need for retiming the stream.

12.4.2.9. Descriptors for Inactive Channels

For virtual channels that are not currently active, no service location descriptor should be carried in the bit stream. Other descriptors for inactive virtual channels may, however, be carried by the transport stream.

12.4.3. ATSC Tables

The tables used by ATSC, including those defined by MPEG-2, are shown in Table 12.8. Of the new tables defined by ATSC, the event information table and the extended text table have PIDs assigned by the master guide table. All other ATSC-defined tables use PID 0x1FFB and are distinguished from one another only by the different values of **table_id** assigned.

12.4.3.1. ATSC Use of the Program Map Table

In an ATSC transport stream, the program map table may carry a number of ATSC-defined descriptors in addition to those defined by MPEG-2, including the AC-3

component_name_descriptor	
component_name_string	Multiple string structure

Figure 12.65 Structure of the ATSC component name descriptor.

Table 12.8. ATSC Tables.

Table	Abbreviation	PID	table_id
Program association	PAT	0x0000	0x00
Conditional access	CAT	0x0001	0x01
Program map	PMT	Assigned in PAT	0x02
Master guide	MGT	0x1FFB	0xC7
Virtual channel (terrestrial)	VCT	0x1FFB	0xC8
Virtual channel (cable)	VCT	0x1FFB	0xC9
Rating region	RRT	0x1FFB	0xCA
Event information	EIT	Assigned in MGT	0xCB
Extended text	ETT	Assigned in MGT	0xCC
System time	STT	0x1FFB	0xCD

audio descriptor (to describe audio elementary streams), the caption service descriptor, the content advisory descriptor, and the component name descriptor (to assign names to individual components).

12.4.3.2. ATSC System Time Table

The system time table specifies the current time and date and the status of daylight saving. The **protocol_version**, which currently must be set to zero, is included to facilitate future evolution of the protocol. The **system_time** is specified using the format described in Section 12.4.1.2. The **GPS.UTC_offset** (measured in seconds) can be subtracted from the system time, which is calibrated in GPS time, to obtain a UTC time.

The use of daylight savings in converting between UTC and local time is specified by the **DS_status**, **DS_day_of_month**, and **DS_hour** fields. **DS_status** takes the value 1 when daylight savings time is active, zero otherwise. In months where the status of daylight savings changes, **DS_day_of_month** indicates the date on which this is to occur, and **DS_hour** the time on that day. In other months, **DS_day_of_month** and **DS_hour** are set to zero.

Although not prohibited by the syntax, the system time table would not normally carry any descriptors (Fig. 12.66).

system_time_table_section	
protocol_version	Unsigned integer
system_time	Time and date
GPS.UTC_offset	Unsigned integer
DS_status	flag
DS_day_of_month	Unsigned integer
DS_hour	Unsigned integer
<< descriptors >>	

Figure 12.66 Structure of the ATSC system time table.

EXAMPLE 12.30—System Time Table

The GPS time at 0800 UTC, August 7, 2001, with an offset of 5 s is 681206405. The system time table for this time, with daylight savings active and no change in daylight savings status during August, is shown in Figure 12.67.

system_time_table_section	
protocol_version	0
system_time	681206405
GPS_UTC_offset	5
DS_status	1
DS_day_of_month	0
DS_hour	0

Figure 12.67 Example ATSC system time table. ■

12.4.3.3. ATSC Master Guide Table

The master guide table carries information on all PSIP tables in a transport stream, except for the system time table, using the structure shown in Figure 12.68. Descriptors defined by MPEG-2 and ATSC are not usually carried in the master guide table, even though carriage of descriptors is supported by the syntax. The data of the master guide table begins with a **protocol_version** field, which is intended to enable future evolution of the protocol. For each PSIP table, the master guide table carries the table type, the PID of the transport stream packets in which this table is carried (**table_type_PID**), the current version number of the table (**table_type_version_number**), and the total number of bytes used by this table. Specification of the version number allows a decoder to determine from the master guide table when a new version of a table is transmitted, without having to monitor all PSIP data in the transport stream. Direct specification of the number of bytes used by a table is designed to assist a decoder to manage its memory. The master guide table is not scrambled.

master_guide_table_section	
protocol_version	Unsigned integer
* {	table_type { VCT,ETT-V,EIT0-127,ETT0-127,RRT1-255 }
	table_type_PID Unsigned integer
	table_type_version_number Unsigned integer
	number_bytes Unsigned integer
	<<descriptors>>
<< descriptors >>	

Figure 12.68 Structure of the ATSC master guide table.

EXAMPLE 12.31—System Time Table

A partial example of the ATSC master guide table is shown in Figure 12.69, for which the protocol version is 0. The virtual channel table is located in PID 0x1FFB (as required for this table), its current version number is 5 and length 354 bytes. The rating region table for region

1 is located in PID 0x1FFB (as required), with version number 3 and size 74 bytes. EIT-0 is located in PID 0x0543, with version number 45 and size 742 bytes, whereas EIT-1 is located in PID 0x523, with version number 47 and size 890 bytes.

master_guide_table_section	
protocol_version	0
table_type	VCT
table_type_PID	0x1FFB
table_type_version_number	5
number_bytes	354
table_type	RRT-1
table_type_PID	0x1FFB
table_type_version_number	3
number_bytes	74
table_type	EIT-0
table_type_PID	0x0543
table_type_version_number	45
number_bytes	742
table_type	EIT-1
table_type_PID	0x523
table_type_version_number	47
number_bytes	890

Figure 12.69 Example ATSC master guide table. ■

12.4.3.4. ATSC Virtual Channel Table

ATSC specifies two versions of the virtual channel table, the first for terrestrial broadcast and the second for cable. Only the terrestrial virtual channel table is described here.

The virtual channel table uses the **protocol_version** field in the same way as the master guide table, allowing future evolution of the protocol. For each virtual channel described, the major channel number and minor channel number are specified, followed by the **modulation_mode**, which may be analog, ATSC (8 or 16 VSB), two modes designed for transmission over a cable network, or specified in private data. The carrier frequency is transmitted as a multiple of 10 Hz, followed by the MPEG-2 transport stream id and program number associated with the virtual channel. The presence of any extended text description is flagged by the **ETM_location**, whereas the use of conditional access on any component of a virtual channel is indicated by setting **access_controlled** to 1. A virtual channel may be hidden from viewing by entering its channel number using the **hidden** flag, or it may be hidden from an EPG using the **hide_guide** flag. This latter case may be useful for NVD services, where the EPG may show a single virtual channel. The **service_type** specifies whether the virtual channel is an analog transmission, ATSC television, or audio or ATSC data broadcast service. The **source_id** indicates the programming source of the virtual channel and has provision for

virtual_channel_table_section

protocol_version	Unsigned integer
major_channel_number	Unsigned integer
minor_channel_number	Unsigned integer
modulation_mode	{Analog,ATSC(8VSB),ATSC(16VSB),Cable Mode 1,Cable Mode 2,Private}
carrier_frequency	Unsigned integer (MHz)
channel_TSID	Unsigned integer (MPEG-2 transport_stream_id)
program_number	Unsigned integer (MPEG-2 program number)
* ETM_location	{No ETM,ETM in PSIP VC,ETM in channel_TSID}Q
access_controlled	flag
hidden	flag
hide_guide	flag
service_type	{analog, ATSC television,ATSC audio,ATSC data}
source_id	Unsigned integer
<<descriptors>>	
<< additional descriptors >>	

Figure 12.70 Structure of the ATSC virtual channel table.

registration of values, enabling the unique identification of a content provider. No two virtual channels within any transport stream for which the virtual channel table carries data should have the same value of **source_id**, unless one virtual channel is a time-shifted version of the other. Descriptors are usually placed in the virtual channel table to describe one virtual channel, although there is provision in the syntax for additional descriptors that are not specific to one virtual channel (Fig. 12.70).

Descriptors typically carried in the virtual channel table are the extended channel name descriptor, the service location descriptor, and the time-shifted service descriptor. The virtual channel table is not scrambled.

EXAMPLE 12.32—Virtual Channel Table

An example virtual channel table is shown in Figure 12.71. Each of the three virtual channels has major channel number 12, has no extended text message (ETM), is not access controlled, is not hidden, and is not hidden in the EPG. The first and third virtual channels are ATSC digital television services, using 8-VSB modulation, with transport_stream_id 0x453D and transmission frequency 540.32 MHz. The MPEG-2 program number (which is used in the PMT) of the first virtual channel is 4 and that of the third virtual channel is 6. The analog virtual channel is assigned transport stream id and program number 0xFFFF.

12.4.3.5. ATSC Rating Region Table

The rating region table carries information on program rating for up to 255 distinct geographic regions. The purpose of the table is to define the ratings system used; the actual rating of a particular event is carried by the content rating descriptor in the event information table or in the program map table.

The region to which a section of the rating region table refers is determined by the value of the **rating_region** field. This value is also used to refer to this region

virtual_channel_table_section	
protocol_version	0
major_channel_number	12
minor_channel_number	0
modulation_mode	ATSC(8VSB)
carrier_frequency	800.31 MHz
channel_TSID	0x453D
program_number	0x4
ETM_location	No ETM
access_controlled	0
hidden	0
hide_guide	0
service_type	ATSC television
source_id	0x1242
<<descriptors>>	
major_channel_number	12
minor_channel_number	1
modulation_mode	Analog
carrier_frequency	205.25 MHz
channel_TSID	0xFFFF
program_number	0xFFFF
ETM_location	No ETM
access_controlled	0
hidden	0
hide_guide	0
service_type	Analog
source_id	0x1242
<<descriptors>>	
major_channel_number	12
minor_channel_number	2
modulation_mode	ATSC(8VSB)
carrier_frequency	800.31 MHz
channel_TSID	0x453D
program_number	0x6
ETM_location	No ETM
access_controlled	0
hidden	0
hide_guide	0
service_type	ATSC television
source_id	0x1242
<<descriptors>>	

Figure 12.71 Example ATSC virtual channel table (without descriptors). ■

in other tables, such as the master guide table. The text name of the rating region is carried in the **rating_region_name_text** field. For each rating dimension (such as “violence,” “nudity,” or “language”) to be defined, **dimension_name_text** specifies the name of the dimension. Dimensions for which a graduated scale of increasing content is to be defined are identified by setting **graduated_scale** to 1. For each value of the dimension, **abbrev_rating_value_text** contains an abbreviated version of the name of the rating value, whereas **rating_value_text** contains the full text of the name (Fig. 12.72).

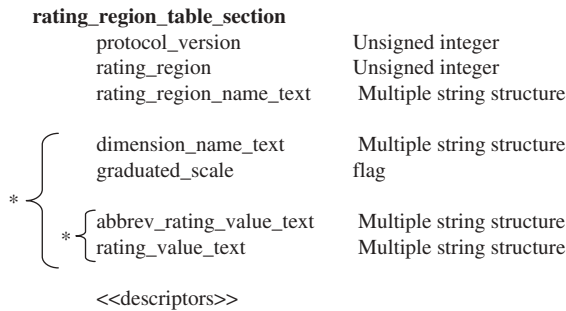


Figure 12.72 Structure of the ATSC rating region table.

The rating region table is not scrambled and does not usually carry ATSC-defined descriptors.

EXAMPLE 12.33—Rating Region Table

An example rating region table, which partially defines the television program classification system used in Australia, is shown in Figure 12.73. The **region_rating_name** shows the region as being Australia. Two dimensions are defined: “language” and “other.” The language dimension is a graduated scale, using four levels: “some coarse language,” “frequent coarse

rating_region_table_section		
protocol_version		0
rating_region		1
rating_region_name_text		“Australia”
dimension_name_text		“Language”
graduated_scale		1
abbrev_rating_value_text		“SCL”
rating_value_text		“Some coarse language”
abbrev_rating_value_text		“FCL”
rating_value_text		“Frequent coarse language”
abbrev_rating_value_text		“VCL”
rating_value_text		“Very coarse language”
abbrev_rating_value_text		“FVCL”
rating_value_text		“Frequent very coarse language”
dimension_name_text		“Other”
graduated_scale		0
abbrev_rating_value_text		“A”
rating_value_text		“Adult themes”
abbrev_rating_value_text		“N”
rating_value_text		“Nudity”

Figure 12.73 Example ATSC rating region table.

language,” “very coarse language,” and “frequent very coarse language.” The other dimension is not a graduated scale, but instead indicates the presence of particular features, such as “nudity” or “adult themes.” For each of these rating values, both the full text and an abbreviation are included. ■

12.4.3.6. ATSC Event Information Table

The event information table is used to carry information on the contents of virtual channels. Each section of the event information table carries information on one virtual channel, which is identified by its **source_id** (which can be used to link to the corresponding information in the virtual channel table).

The structure of the event information table is shown in Figure 12.74. Following the **source_id** and **protocol_version** fields, one or more events on the virtual channel are described. Each event is assigned a unique identifier (**event_id**), and the start time, length, title, and the location of any ETM event are specified.

Each event information table describes events in a 3-h window. Each 3-h window begins at 0000, 0300, 0600, 0900, 1200, 1500, 1800, or 2100 UTC. EIT-0 always specifies current events, EIT-1 specifies events in the next 3-h window, and the following event information tables specify events in future 3-h windows. For example, if the current time is 0500, EIT-0 contains information on events in the period 0300–0600, EIT-1 on 0600–0900, EIT-2 on 0900–1200, and so on. A minimum of four event information tables must be carried, describing events in the current 3-h period and the following three 3-h periods.

ATSC descriptors commonly carried in the event information table are the AC-3 audio descriptor, the caption service descriptor, and the content advisory descriptor. The event information table is not scrambled.

EXAMPLE 12.34—Event Information Table

An example event information table for EIT-0 valid at 0930 UTC, January 26, 2000, is shown in Figure 12.75.

The first event, whose title is “Who wants to be a millionaire,” has **event_id** 0x0425, starts at 0830 (coded as 0x25B97308), and is 1-h long. The second event, whose **event_id** is 0x0625, starts at 0930 (0x25B98118), is 90 min long, has extended text in the virtual channel carrying the event and has the title “Friends.” The third event (0x0429) starts at 1100 (0x25B99630), is 60 min long, and has the title “The Late Show.”

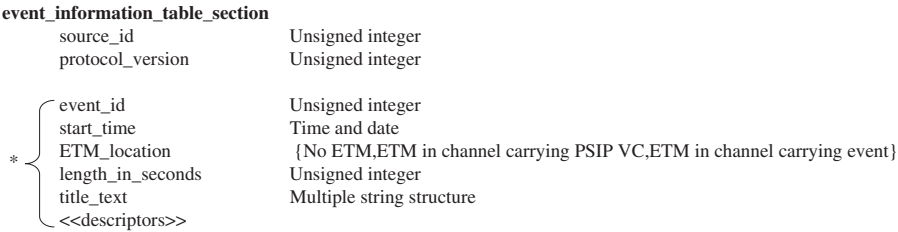


Figure 12.74 Structure of the ATSC event information table.

event_information_table_section	
source_id	0x4352
protocol_version	0
event_id	0x0425
start_time	0x25B97308
ETM_location	No ETM
length_in_seconds	3600
title_text	“Who wants to be a millionaire”
<<descriptors>>	
event_id	0x0625
start_time	0x25B98118
ETM_location	ETM in channel carrying event
length_in_seconds	5400
title_text	“Friends”
<<descriptors>>	
event_id	0x0429
start_time	0x25B99630
ETM_location	No ETM
length_in_seconds	3600
title_text	“The Late Show”
<<descriptors>>	

Figure 12.75 Example ATSC event information table. ■

12.4.3.7. ATSC Extended Text Table

The extended text table is used to carry longer text descriptions of events and virtual channels, and contains one multiple string structure (Fig. 12.76). For an event, the **ETM_id** (32 bits) combines the **source_id** (16 bits) of the virtual channel and the **event_id** (16 bits) of the event. For a virtual channel, the **source_id** is combined with 16 zero bits.

The extended text table is not scrambled.

12.5. DVB SI AND ATSC PSIP INTEROPERABILITY

It may sometimes be desirable to create a transport stream that can be decoded by either a DVB or an ATSC decoder. This is most likely to occur when a transport stream is to be distributed into a number of countries.

In practice, interoperability in this respect does not mean that the DVB receiver decodes ATSC PSIP or that the ATSC receiver decodes DVB SI. What is meant is that the transport stream carries both DVB SI and ATSC PSIP in a manner such that no conflicts between the two systems arise, which imposes a number of restrictions

extended_text_table_section	
protocol_version	Unsigned integer
ETM_id	Unsigned integer
extended_text_message	Multiple string structure

Figure 12.76 Structure of the extended text table.

on the structure of the transport stream to ensure that conflicts between DVB and ATSC do not arise [10].

The restrictions imposed by this requirement are in the areas of allocation of PIDs to transport stream packets, and the use of values of **table_id** or **descriptor_tag** assigned by either DVB or ATSC.

12.5.1. PIDs

PIDs in the range 0x1FFB–0x1FFD are reserved for use by ATSC (e.g., 0x1FFB is used by PSIP) and should not be used by DVB. Similarly, PIDs in the range 0x0010 through 0x0014 are reserved by DVB and should not be used by ATSC. This latter restriction means that ATSC program number 1 should not be used.

12.5.2. Use of table_id

The ranges of values reserved by DVB (0x40 through 0x7F) and ATSC (0xC0 through 0xFE) should not be used for user-private data.

12.5.3. Use of descriptor_tag

Descriptor tag values from 0x80 to 0xAF are used or reserved for use in the ATSC specification, whereas 0x40 through 0x7F are reserved in the DVB specification. User-private descriptors should use descriptor tag values outside this range. The range 0xB0 through 0xFE can be safely used for user-private descriptors, the use of which is controlled by the MPEG-2 registration descriptor.

12.6. CONCLUSION

DVB SI and ATSC PSIP extend the basic functionality of the MPEG-2 PSI, providing a range of additional information about the programs carried in one or more transport streams. This additional information supports applications such as electronic program guides, assists the viewer in navigating between different programs, and allows the appearance of service offerings to be made independent of the physical structure of MPEG-2 transport streams.

PROBLEMS

- 12.1** Using the format of Figure 12.40, construct a network information table for a satellite distribution system with two channels on one satellite with orbital position 130.5°E, **original_network_id** 0x1010, and network identifier 0x0432, with the following characteristics:

- Channel 1 is transmitted with **transport_stream_id** 0x0786, at a frequency of 3587 MHz, with a channel bandwidth of 10 MHz, vertical polarization, a symbol rate of 10 Msym/s, and an inner coding rate of 2/3.
- Channel 2 is transmitted with **transport_stream_id** 0x0765, at a frequency of 3456 MHz, with a symbol rate of 20 Msym/s, a bandwidth of 20 MHz, left circular polarization, and an inner coding rate of 1/2.

12.2 Answer the following questions, using the information in the example network information table shown in Figure 12.77.

network_information_section-actual_network	
network_id	0x0163
network_name_descriptor	
char	“BBC”
transport_stream_id	0x0234
original_network_id	0x0001
terrestrial_delivery_system_descriptor	
center frequency	500.5 MHz
bandwidth	7 MHz
constellation	QPSK
hierarchy information	Non-hierarchical
code_rate-HP_stream	1/2
code_rate-LP_stream	1/2
guard_interval	1/4
transmission mode	2k
other_frequency_flag	None
transport_stream_id	0x0239
original_network_id	0x0001
terrestrial_delivery_system_descriptor	
center frequency	630.0 MHz
bandwidth	7 MHz
constellation	64-QAM
hierarchy information	Non-hierarchical
code_rate-HP_stream	7/8
code_rate-LP_stream	7/8
guard_interval	1/32
transmission mode	8k
other_frequency_flag	None

Figure 12.77 Example network information table for Question 12.2.

- How many transport streams are carried by this network?
- Which fields uniquely identify each transport stream?
- For each descriptor, whether it refers to one transport stream or to the whole network?

12.3 A transport stream carries five programs with no conditional access. If each table consists of exactly one section, and each section occupies one transport stream packet, calculate the overhead in bytes per second in the following cases:

- (a) DVB SI (consisting of program association table, program map table, network information table, bouquet association table, network information table, service description table, event information table (present/following), and time offset table) is broadcast twice per second.

- (b) ATSC PSIP (consisting of program association table, program map table, system time table, master guide table, virtual channel table, event information table, and rating region table) is broadcast twice per second.
- (c) Both DVB SI and ATSC PSIP are carried.

12.4 Fill in the blank entries in Table 12.9.

Table 12.9 Corresponding values of date and MJD for Question 12.4.

Date	MJD
January 5, 2001	
September 9, 1999	52035
	51106

12.5 Fill in the blank entries in Table 12.10. The number of days since the reference date can be found using the difference between their values of MJD.

Table 12.10 Corresponding entries of GPS time and date and time for Q 12.5.

Time and date	GPS time
8:45 a.m., August 8, 1995	
3:40 p.m., February 29, 1996	673275360
	545217240

- 12.6 Use the data in Table 12.11, which describes the programs and elementary streams carried in a DVB transport stream, to construct a program association table and program map tables. The transport stream has a maximum bit rate of 20 Mbit/s, **original_network_id** is 0x0100, and **transport_stream_id** is 0x0034. In answering this question, it may be necessary to assign PIDs or other parameters. No conditional access is used in this transport stream.
- 12.7 Construct a bouquet association table based on the data provided in Tables 12.11–12.13.
- 12.8 Construct a service description table based on the data provided in Tables 12.12 and 12.13. All services are running and event information tables are carried in the transport stream for present/following events but not for the schedule.
- 12.9 Construct an event information table for Service 1, using the data in Tables 12.12–12.14.
- 12.10 Is the DVB transport stream defined in Tables 12.11–12.14 compatible with ATSC, in the sense that ATSC PSIP could be added to the transport stream to allow decoding by an ATSC decoder?

Table 12.11 Elementary stream data for Questions 12.6–12.9.

Program number	PID	Elementary stream type	Component name	Elementary stream description
—	0x0056	EPG		EPG for Bouquet 1
—	0x0057	EPG		EPG for Bouquet 2
1	0x0365	MPEG-2 video	BBC1—video	Main profile at main level; 4:2:0; 25 Hz; aligned at slice layer
	0x366	MPEG-2 audio	BBC1—audio	English language; constant rate; Layer 2; ID = 1; free_format_flag = 0
	0x375	DVB subtitle	BBC1—subtitles	English (hearing impaired, composition_page_id = 1, ancillary_page_id = 10) and French language (normal, composition_page_id = 11, ancillary_page_id = 43)
	0x377	DVB teletext	BBC1—teletext	English (initial page in magazine 0, page 10) and French language (initial page in magazine 1, page 10)
	0x0365	PCR		Elementary stream carrying PCR for this program
2	0x0385	MPEG-2 video	BBC2—video	Main profile at main level; 4:2:0; 30 Hz; aligned at picture layer
	0x386	MPEG-2 audio	BBC2—English audio	English language; constant rate; Layer 2; ID = 1; free_format_flag = 0
	0x387	MPEG-2 audio	BBC2—French audio	French language; constant rate; Layer 2; ID = 1; free_format_flag = 0
	0x0385	PCR		Elementary stream carrying PCR for this program
3	0x0435	MPEG-2 video	Sky—video	Main profile at main level; 4:2:0; 30 Hz; aligned at picture layer
	0x436	MPEG-2 audio	Sky—English audio	English language; constant rate; Layer 2; ID = 1; free_format_flag = 0

Table 12.11 (Continued)

Program number	PID	Elementary stream type	Component name	Elementary stream description
	0x437	MPEG-2 audio	Sky—German audio	German language; constant rate; Layer 2; ID = 1; free_format_flag = 0
	0x0435	PCR		Elementary stream carrying PCR for this program

Table 12.12 Service description information associated with components defined in Table 12.11.

Service identifier	Service name	Service provider
1	BBC-1	British Broadcasting Corporation
2	BBC-2	British Broadcasting Corporation
3	Sky Sports	Sky Television

Table 12.13 Bouquet information associated with data in Tables 12.12 and 12.13.

Bouquet identifier	Bouquet name	Services	Country availability	EPG PID
1	BBC	1, 2	UK, France	0x0056
2	News	1, 3	UK, Germany	0x0057

Table 12.14 Data for event information table in Questions 12.9 and 12.14.

Event identifier	Start time	Duration (min)	Running status	Event name	Description
0x23	7:00 p.m., January 1, 2000	90	Running	Premier league football	Manchester United versus Arsenal
0x6B	8:30 p.m., January 1, 2000	30	About to start	Nightly news	National and international news from the BBC

- 12.11** Use the data in Table 12.15, which describes the programs and elementary streams carried in an ATSC transport stream, to construct a program association table and program map tables. The transport stream has a maximum bit rate of 20 Mbit/s, and **transport_stream_id** is 0x0034. In answering this question, follow the ATSC program number convention. No conditional access is used in this transport stream.
- 12.12** Using the data in Tables 12.15 and 12.16, construct an ATSC virtual channel table for a station using Channel 7 for its preexisting NTSC analog service and Channel 14 for its digital service. There is no extended text for these programs.
- 12.13** Construct an ATSC master guide table for the services in Questions 12.9–12.11. Where necessary, assign PIDs for tables that have not yet been defined. All tables have version number 7 and size 875 bytes.
- 12.14** Use the data in Table 12.14 to construct an ATSC event information table for the transport stream defined in Tables 12.15 and 12.16.

Table 12.15 Elementary stream data for Questions 12.11–12.14.

Program number	PID	Elementary stream type	Component name	Elementary stream description
2	0x0021	MPEG-2 video	NBC-video	Main profile at main level; 4:2:0; 30 Hz; aligned at slice layer, with English (Line 21) closed captioning in Field 1, designed for beginner readers.
	0x0024	AC-3 audio	NBC-audio	English language; 44.1 kHz, 320 kbit/s, two channels.
3	0x0031	MPEG-2 video	CBS-audio	Main profile at main level; 4:2:0; 30 Hz; aligned at picture layer; with German (ATVCC) closed captioning using caption service number 3.
	0x0034	AC-3 audio	CBS-English audio	English language; 48 kHz, 320 kbit/s, two channels.
	0x0038	AC-3 audio	CBS-French audio	French language; 44.1 kHz, 320 kbit/s, two channels.
4	0x0041	MPEG-2 video	ABC-video	Main profile at main level; 4:2:0; 30 Hz; aligned at picture layer; with English (Line 21) closed captioning in Field 2.
	0x0044	AC-3 audio	ABC-English audio	English language; 44.1 kHz, 320 kbit/s, two channels.
	0x0047	AC-3 audio	ABC-German audio	German language; 44.1 kHz, 640 kbit/s, two channels.

Table 12.16 Data for virtual channel table for Question 12.12.

Minor channel number	Program number	Service type	Carrier frequency (MHz)	Source identifier
1	—	Analog	175.25	0x5345
2	2	ATSC television	470.310	0x5345
3	3	ATSC television	470.310	0x5345
4	4	ATSC television	470.310	0x5345

12.15 Is the transport stream described by Question 12.10 capable of carrying DVB-SI and thereby being decodable by a DVB decoder?

12.16 Construct an ATSC rating region table from the data in Table 12.17.

Table 12.17 Rating region data for Question 12.15.

Dimension name	Graduated scale	Rating value text	Abbreviated rating value text
Violence	1	Some violence	SV
		Frequent violence	FV
		Strong violence	StV
Sex	1	Sexual reference	SR
		Sex scenes	SS
		Strong sex scenes	StS
Other	0	Medical procedures	M
		Horror	H

12.17 Construct an ATSC system time table specifying the time as 0020 UTC on 4th March, 1999. The offset between UTC and GPS is 14 s, and daylight savings is inactive. The next change of daylight savings status is to occur on October 1, 1999, at 2 a.m.

12.18 Construct a DVB time offset table for New Zealand (which has only one time zone), specifying that the current time is 8 p.m., January 3, 1998 (local time), with an offset of +13 h from UTC, and also specifying that this offset will change to +12 h at 2 a.m. (local time), April 1, 1998.

MATLAB EXERCISE 12.1

ATSC and DVB make use of different time and date representations.

The aim of this exercise is to implement MATLAB functions to calculate the modified Julian date and GPS time, and to use these to calculate the days of week.

1. Using the formula provided in Section 12.3.1, implement a MATLAB function to calculate the modified Julian date, given the year, month, and day of month.

2. Write a MATLAB function to translate an array in the format of the output of the MATLAB clock() function into the GPS time used by ATSC. (Hint: The number of days since January 6, 1980, can be easily found using the modified Julian date function from Part 1 of this exercise.)
3. Implement a function to calculate the days of week (i.e., Sunday, Monday, etc.) from the modified Julian date. (Hint: Start by picking a reference date for which the day of week is known. Find the modified Julian date of this reference date and the date for which the day of week is required. The required day of week can be calculated from the difference in the modified Julian dates.

REFERENCES

1. DVB specifications relating to SI are
 - (a) Digital Video Broadcasting (DVB); *Specification for Service Information (SI) in DVB Systems*, EN 300 468, Sophia Antipolis: ETSI, 1998.
 - (b) Digital Video Broadcasting (DVB); *Guidelines on Implementation and Usage of Service Information (SI)*, ETR 211, Sophia Antipolis: ETSI, 1997.
 - (c) Digital Video Broadcasting (DVB); *Allocation of Service Information (SI) Codes for DVB Systems*, ETR 162, Sophia Antipolis: ETSI, 1995.
2. The formal definitions for character sets used in DVB-SI are contained in
 - (a) ISO 6937, *Information technology—coded graphic character set for text communication—Latin alphabet*.
 - (b) ISO 8859, *Information processing—8-bit single-byte coded graphic character sets, Latin alphabets*.
3. ISO 3166, *Codes for the representations of names of countries*.
4. ISO 639, *Code for the representation of names of languages*.
5. Digital Video Broadcasting (DVB); *Specification for Service Information (SI) in DVB Systems*, EN 300 468, Sophia Antipolis: ETSI, 1998, Annex C, pp. 71–72.
6. Digital broadcasting systems for television, sound and data services; *Allocation of Service Information (SI) Codes for Digital Video Broadcasting (DVB) Systems*, ETR 162, Sophia Antipolis: ETSI, 1995.
7. Digital Video Broadcasting (DVB); *Guidelines on implementation and usage of Service Information (SI)*, ETR 211, Sophia Antipolis: ETSI, 1997, Section 4.4, pp. 27–28.
8. ATSC Standard A/65A, *Program and system information protocol for terrestrial broadcast and cable*, Advanced Television Systems Committee, May 2000.
9. This descriptor is specified in ATSC Standard A/58, *Digital audio compression standard (AC-3)*, Advanced Television Systems Committee, December 1995.
10. This issue is discussed further in
 - (a) Digital Video Broadcasting (DVB); *Guidelines on implementation and usage of Service Information (SI)*, ETR 211, Sophia Antipolis: ETSI, 1997, Annex A.
 - (b) ATSC Standard A/58, *Harmonization with DVB SI in the use of the ATSC Digital Television Standard*, Advanced Television Systems Committee, September 1996.
 - (c) AS 4599-1999, *Digital television—terrestrial broadcasting—characteristics of digital terrestrial television transmissions*, Homebush NSW: Standards Australia, 1999, Annex A to Section 7.